

Lab 08 - Subnet Masks and Segmentation



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Lattes ORCID SciProfiles Scopus ResearcherID substationworm LFFreitasGutierrez

Tasks

- **1** Verify the IP and MAC addresses of the pc1 workstation.
- **2** Identify the subnet mask associated with the pc1 workstation.
- **3** Repeat the previous steps for the workstations pc2, pc3, pc4, pc5, and pc6. Compare the addressing schemes and determine the underlying segmentation logic.
- **4** From pc1, use the ping command to test connectivity with pc2 and pc3. Provide an interpretation of the results.
- **5** From pc1, attempt to ping pc4. Evaluate the outcome.
- **6** From pc5, attempt to ping pc6. Analyze the results.
- **7** Execute the ipcalc command from any container to calculate the network information relevant to this study.
 - *Identify the network address, broadcast address, and number of available hosts.*
- **8** On pc2, run traceroute <IP address of pc4> and explain the observed routing path.
- **9** On pc2, execute traceroute <IP address of pc3> and interpret the resulting path.
- **10** From pc3, perform a network discovery scan using nmap -sn <network_address/24> for each network analyzed in this study. Identify the active hosts detected.

Tools

- The following tools are available on the pc1, pc2, pc3, pc4, pc5, and pc6 workstations for completing OTLab 08: ifconfig, ipcalc, masscan, nmap, ping, and traceroute.

Nomenclature

- IP: Internet protocol.
- MAC: Media access control.